

Lecture Notes on

Unit 02 Differential Equations

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Within the contents of this is my study guide/ personal notes to MAP2302 which was taken in the Spring of 2026 at Palm Beach State (PBS) Community College in Lake Worth, Florida under Professor Tamara Johns. This examination will be taken on Friday, March 6th 2026 at blah blah blah from the hours of 3pm - 9pm.

This chapter predominantly covers the following topics (General):

1. Lesson 7 Higher-Order Linear Differential Equations
2. Lesson 8 Higher-Order Differential Equations / Reduction of Order
3. Lesson 9 Homogenous Linear Equations with Constant Coefficients
4. Lesson 10 Undetermined Coefficients - Superposition Approach
5. Lesson 11 Undetermined Coefficients - Annihilator Approach

The main revisions made in this study guide versus Unit 01 will be the organization and the utilization of some sort of Appendix to be able to keep track of misc things/ concepts (int. by parts shortcuts to save time, intermediate trig, etc.).

I also want to make use of subsubsections in this paper, so hopefully that goes well. I think that the organizational structure of a toc that utilizes subsubsections looks really clean, as often done by professor David Tong.

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1 Higher Order Linear Differential Equations

1.1 Introduction to Higher Order Linear ODEs

1.1.1 General Form and Terminology

1.1.2 Existence and Uniqueness of Solutions

1.1.3 The Superposition Principle

1.2 The Structure of Solutions to Homogeneous Equations

1.2.1 Linear Independence and Solution Sets

1.2.2 Fundamental Sets of Solutions

1.3 The Wronskian and Linear Independence

1.3.1 Definition and Computation of the Wronskian

1.3.2 Abel's Theorem

1.3.3 Using the Wronskian to Verify Independence

2 Reduction of Order

2.1 Introduction to Reduction of Order

2.1.1 Motivation: When One Solution Is Known

2.1.2 Deriving the Reduced Equation via Substitution

2.2 Homogeneous Equations

2.2.1 Worked Example: Finding a Second Independent Solution

2.2.2 Verification via the Wronskian

2.3 Non-Homogeneous Equations

2.3.1 Adapting Reduction of Order for Non-Homogeneous Cases

2.3.2 Worked Example: Second-Order Non-Homogeneous Equation

3 Homogeneous Linear Equations with Constant Coefficients

3.1 Introduction and the Characteristic Equation

3.1.1 Deriving the Characteristic (Auxiliary) Equation

3.1.2 General Solution Framework

3.2 Case I: Distinct Real Roots

3.2.1 Solution Form and General Solution